# Pharmacy Sales and Drug Analysis

Medicine usage trends and revenue patterns, January to December 2025

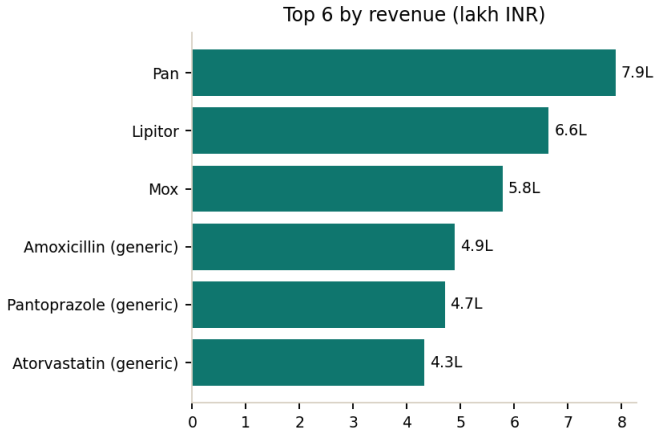
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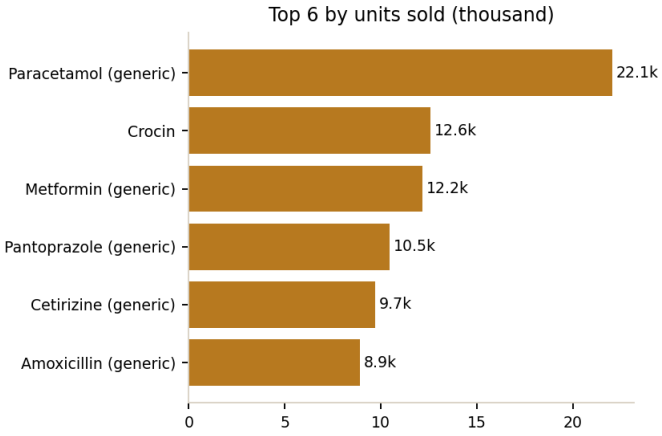
**Objective.** Study pharmacy sales to find top-selling medicines, monthly revenue growth and the difference between branded and generic medicines. Tools: Python (pandas, matplotlib). Dataset: a fictional sample of 144 records (12 months x 12 products) created for practice; it had no missing or duplicate values.

| INR 49.2 lakh | 111,446 | Pan | Paracetamol* | +23.8% |
|---|---|---|---|---|
| Total revenue | Units sold | Top by revenue | Top by units | Dec vs Jan revenue |

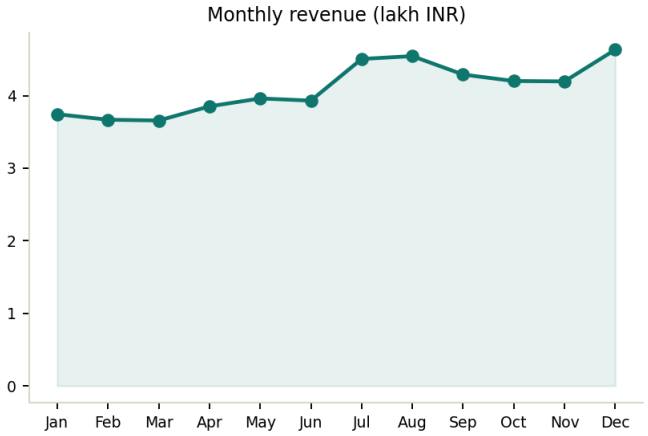
## 1. TOP-SELLING MEDICINES

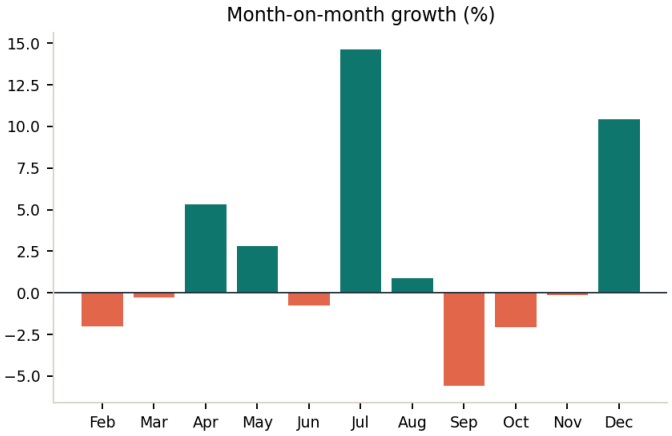
| Rank | Top 5 by revenue | Revenue (INR) | Top 5 by units | Units |
|---|---|---|---|---|
| 1 | Pan | 788,880 | Paracetamol (generic) | 22,051 |
| 2 | Lipitor | 664,320 | Crocin | 12,570 |
| 3 | Mox | 578,930 | Metformin (generic) | 12,181 |
| 4 | Amoxicillin (generic) | 489,225 | Pantoprazole (generic) | 10,456 |
| 5 | Pantoprazole (generic) | 470,520 | Cetirizine (generic) | 9,686 |




## 2. MONTHLY SALES AND GROWTH




| Month | Revenue (INR) | Growth vs previous month | Month | Revenue (INR) | Growth vs previous month |
|---|---|---|---|---|---|
| January | 374,410 | - | July | 450,601 | +14.6% |
| February | 366,850 | -2.0% | August | 454,497 | +0.9% |
| March | 365,756 | -0.3% | September | 429,113 | -5.6% |
| April | 385,259 | +5.3% | October | 420,184 | -2.1% |
| May | 396,154 | +2.8% | November | 419,662 | -0.1% |
| June | 393,148 | -0.8% | December | 463,395 | +10.4% |

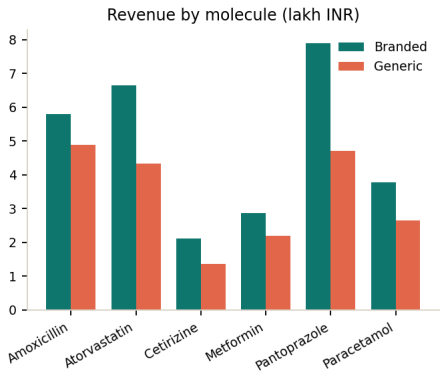
## 3. BRANDED VS GENERIC MEDICINES

| Measure | Branded | Generic |
|---|---|---|
| Revenue (INR) | 2,906,850 | 2,012,179 |
| Revenue share | 59.1% | 40.9% |
| Units sold | 40,961 | 70,485 |
| Units share | 36.8% | 63.2% |
| Average price per unit (INR) | 71.0 | 28.5 |
| Growth in monthly revenue, Dec vs Jan | +21.8% | +26.8% |

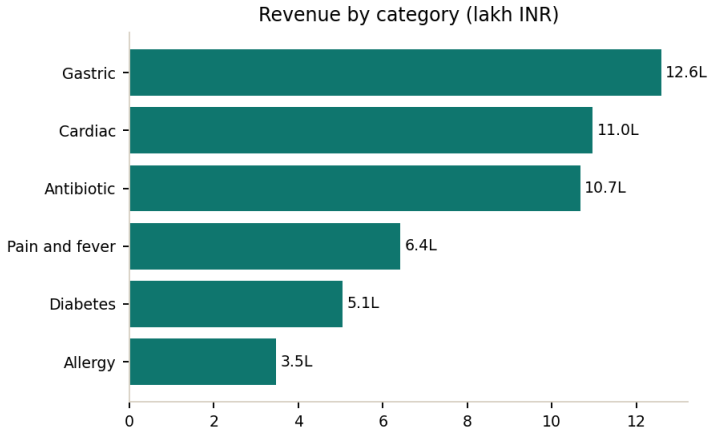


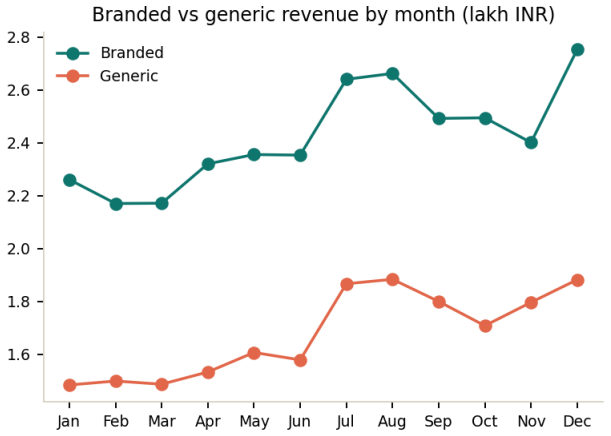







## 4. CATEGORY PERFORMANCE AND TREND




## 5. KEY FINDINGS

• **Top sellers:** Pan earns the most revenue (INR 788,880) because of its higher price, while Paracetamol (generic) sells the most units (22,051). The best seller by units is not the best by revenue.
• **Monthly trend:** Revenue grew from INR 374,410 in January to INR 463,395 in December (+23.8%), averaging +2.1% a month. The best month was December and the weakest was March.
• **Seasonality:** Some drugs peak in particular seasons (amoxicillin in Jul, cetirizine in Nov, paracetamol in Dec), which explains the jumps in July and December. Month-to-month growth therefore mixes real growth with seasonal demand.
• **Branded vs generic:** Generics sold 63% of units but earned 41% of revenue; branded medicines sold 37% of units and earned 59%. The average branded unit brings in INR 71 against INR 28 for generics. Generics outsold branded in every molecule.
• **Categories:** Gastric, Cardiac and Antibiotic medicines bring in the most revenue.
• **Suggestions:** Stock seasonal medicines ahead of their peak months; keep generics available because they drive volume and are affordable for patients; track margins (not in this dataset) before judging which type is more profitable.
• **Limitation:** One year of fictional data from a single pharmacy. Revenue is not profit.

Fictional data for internship practice only. * = generic version. Files: pharmacy_sales_2025.csv, pharmacy_analysis.py.